

CPT 311

COCOTAMS

Crop Production (TARO AND TAMNIA)

(3)

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ORIGIN AND DISTRIBUTION

In many parts of Africa, the term 'Cocoyams' is used to refer collectively to members of the genus Colocasia and the genus Xanthosoma which are used for food. When referred to separately Colocasia species are called 'taro' while Xanthosoma species are called 'tamnia'. Of course, other common names exist for these crops in various parts of the world.

Taro, originated in South Central Asia and was first cultivated in that region. From there it first spread to the Pacific Islands and later to the Mediterranean area and West Africa. From West Africa it was introduced to the West Indies. Today it is grown in nearly all parts of the humid tropics.

Tamnia consists of several species of Xanthosoma with X. sagittifolium being the most important. It originated in Central and South America and was first cultivated there. From there it spread to South-East Asia and the Pacific Islands. It was introduced to Africa in the nineteenth century, much later than taro. Partly for this reason, taro may be called 'Cocoyam' while tamnia is called 'new Cocoyam'.

Today, more than three-quarters of world Cocoyam production comes from Africa. The rest comes from Asia and Oceania. Nigeria, Ghana and Japan are the world's leading producers.

Bony of cocoyams.

Cocoyams are herbaceous plants. Each plant has a more or less spherical corm which lies just beneath the soil. The apex of the corm is the main growing point of the shoot. This growing point lies close to the soil level. Leaves arise from the upper part of the corm, while the fibrous roots extend from practically all parts of the corm. ^{Cornels and daughter corms} These roots arise laterally from the main corm. Tannia is generally a larger and more vigorous plant than taro.

Each corm is perennating structure, densely packed with starch. Leaf scars are prominent on it, and numerous axillary buds (eyes) are present on the body of the corm. Cornels which arise laterally from the corm are in fact stem tubers. They have a terminal bud at their distal end, scale leaves on their surface, and numerous buds in the axils of the scale leaves. Cornels are usually cylindrical or flask-shaped, and each corm may bear numerous cornels at different ages. For taro, ~~both~~ corns and cornels are used for food, but for tannia, preference is given to cornels for consumption.

The aerial portion of cocoyams consists mainly of the leaves. Each leaf is attached to the corm by means of a long petiole which is flared out at the end where it ~~attaches~~ ^{attaches} to the corm. Large air spaces exist in the petiole and in the major veins of the lamina i.e. they ~~are~~ have spongy texture.

^{In a deep indentation which divides the base of lamina into two lobes. The petiole is}
attached at the junction of the margins of these two lobes. From the point of attachment of the lamina, a thick mid-rib runs to the apex of the leaf, while two large veins run to the basal lobes.

For form, the leaf is more circular in shape and the basal lamina lobes are less sharply demarcated than for *taninia*. Moreover, the attachment of the petiole is somewhere in the middle of the lamina and not at the margin where the two lobes meet.

Natural flowering in *Coccyzus* is relatively rare. The inflorescence is a spadix, and each plant may produce several inflorescences one after the other. Each inflorescence consists of a stout peduncle, a large yellowish bract called the spathe, and the spadix which consists of a central axis to which numerous small flowers are attached. The flowers on the spadix are unisexual, the female flowers occurring at the basal portion of the spadix, and the male flowers occurring at the distal part.

Each female flower consists of an ovary with a sessile stigma, while each male flower consists of 2-6 stamens which are fused together. The flowers are scented and pollination is probably by insects. The fruit is a berry. The seed has a hard testa, contains endosperm in addition to the embryo, and germinates with difficulty. Although natural flowering is rare in *Coccyzus*,

flowering occurs readily when the plants are treated with gibberellic acid. This substance is now being used by cocoanut breeders to cause frequent flowering so that desired hybridization can take place.

ECOLOGICAL REQUIREMENTS OF COCONUTS

Cocoanuts have a high moisture requirement and grow best where 2000 mm or more of water can be provided each year through rainfall or irrigation. Moisture stress during the season results in mis-shapen coconuts of poor quality.

Cocoanuts are a warm-weather crop and do best when temperatures are above 21°C . They perform poorly if temperatures are low. It appears that cocoanuts are able to tolerate shady conditions better than most other crops. Taro is able to tolerate waterlogged conditions but for tannin, a well-drained soil is essential. Cocoanuts prefer a soil pH of 5.5-6.5 and are able to tolerate saline soils.

LAND PREPARATION

For the cultivation of tannin, and the non-flooded cultivation of taro, the land is prepared in the same way as for the other tuber crops. Planting on the flat, ridged, or mounds are the common practices. Harvesting is easier on ridged and mounds than on the flat.

For the flooded cultivation of taro, elaborate land preparation may be required to obtain a

~~Pounded.~~ Land preparation in these circumstances may be carried out while the field is wet; sometimes the field is deliberately paddled to aid water retention.

PLANTING MATERIALS

Because of the profusion of buds on the corm and cormels, Cocoloyams can be conveniently established from intact corms, intact cormels, or pieces of corms and cormels. The best planting material is a piece consisting of the apical portion of the corm with the lower 20 cm or so of the petioles still attached. Such a planting piece is called a 'huli' in Hwarai, where it is the major type of planting material. It is preferred to the other kinds of planting material because it establishes more quickly and yields higher than the other kinds.

CROP ESTABLISHMENT

Cocoloyams are best planted at the onset of the rains, so that they have the entire rainy season to make their growth.

A spacing of 60 x 60 cm has been recommended for Cocoloyams. In most of tropical Africa today, Cocoloyams are spaced too wide apart. The crop canopy therefore does not close early enough and weeds are a problem, resulting in relatively low yield. These fields could be improved by adopting a more appropriate spacing. As a rule, closer spacing increases corn yield per hectare but decreases corn yield per plant. Where Cocoloyam

are grown as part of an intercropping mixture, their spacing will vary depending on the nature and density of the other crops ~~depending on the nature and density of the other crops~~

Cocoyam Planting pieces should be planted to a depth of about 7 cm in the soil. If hill or suckers are used, the base of the material should be about 7 cm beneath the soil, while the petiole bases lie above the soil. Deep planting is essential to encourage the plant to root deeper and to ensure that the new corn produced is not partially exposed above the soil. Partly mechanized cocoyam planters are available in various parts of the world. They operate essentially by opening up a deep furrow, having the planting piece inserted, and then closing up the furrow. For the flooded cultivation of taro, the crop is planted into 2-5 cm of standing water. The planting piece is inserted by hand into the loose puddled soil. As the base of the plant continues to grow upwards, the water level is raised in stages so that the plant's base remains continually submerged. The water on the field is regularly replenished with fresh water in order to avoid the build-up of root rot.

CROP MANAGMENT

1. Fertilization.

Cocoyams have a high requirement for potassium and calcium. The crop also responds well to nitrogen fertilizer, but excess nitrogen tends to depress yields. The exact quantities and kind of fertilizer depends on the soil type. Generally, more fertilizer is required for flood taro than for non-flooded taro or taro 19. Weed control: Weeds are a problem in cocoyams during

When the corn is ^{left} to produce new generations of cobs which will be harvested later.

Most Cologans grown in Africa are harvested by hand or with hand tools. If the conditions are dry at harvest time, the roots are usually ahead, so that the crop can be lifted up by pulling the withered aerial parts. In all cases care should be taken to avoid wounding the cobs and cobs during harvest.

YIELDS

Yields of Cologans in the world average about 5 tonnes/ha. The highest yields of 37-75 tonnes/ha are recorded for flooded two in Hawaii. In Africa, yields of ~~1000~~ vary from country to country. Egypt, with 31 tonnes/ha, has some of the highest yields. In Niger and Ghana, yields average 5 tonnes/ha. Tanna yields of 25-37 tonnes/ha have been reported in Puerto Rico, but average yields there are about 12-20 tonnes/ha.

STORAGE

For prolonged storage, Cologans should be kept at about 70°C at a relative humidity of 85%. Under these conditions, Cologans can store well for up to four months. If the temperature is much higher, respiration, rotting and sprouting cause storage life to be shortened. If the temperature is much lower, the buds die and the cobs decay. J. D. A. P. T. D.

The early part of the season ^⑥ when the rice is building up, and during the last two or three months when the leaf area is declining. For dry-land (non-flooded) farms and farms, hand tools are the most common method of weed control.

Various herbicides have proved effective in weed control in cocoyams. TCA (5 kg/ha) used with diuron (3 kg/ha) or atrazine (3 kg/ha) can be used for dry-land farms or farms. A pre-emergence application of prometryne (1.2 kg/ha) can also control weed for the first one or two months after planting.

For flooded farms, the very practice of flooding serves as a weed control measure as only a few weed species are able to survive the flooded conditions. Where necessary, herbicides can be introduced in the irrigation water. Nitrofen at 3-6 kg/ha has been used in this way. Alternatively the land can be drained temporarily and any of the herbicides recommended for dry-land farms can be used.

HARVESTING

Cocoyams are ready for harvest after 7-11 months. The leaves may begin to turn yellow at this time, but if growing conditions are still favourable, crop can be yellowing may fail to occur, and

there may be no morphological changes to indicate maturity. However, there is little or no deterioration if the crop is left in the ground for a few weeks.

~~Pro n. harvested in October, and~~

Even with these methods, the crop can keep satisfactorily for only a few weeks.

ECONOMIC IMPORTANCE OF COCOYAMS

- The fresh cocoyam corn is composed of 70-80% water, 20-25% starch and 1.5-3% protein.
- It also contains significant amounts of Vitamin C, thiamin, riboflavin, niacin and carotene.
- The starch grains of taro are small in size and are more easily digested than those of yam, cassava or sweet potato.
- The protein content of taro and tennia is also higher than that of these other tropical tuber crops.
- Fresh cocoyams contain substances (calcium oxalate crystals) which are irritating to the skin and mouth. The irritating effect is removed when the corn or cormels is cooked.
- The corms or cormels of cocoyams are eaten in the same way as yams or sweet potatoes i.e. the same way as yams. In West Africa, boiled, fried, baked or roasted. In West Africa, the boiled cocoyam is sometimes pounded to produce a paste similar to pounded yam, and eaten in the same way.
- Cocoyam flour is one of the few processed forms of cocoyam encountered in Africa. This is made by peeling and slicing fresh or soaked corms and cormels, drying the sliced, and milling the dried slices into flour. It is used, and milling the

Cojolan leaves are used for human consumption in various parts of the world including Nigeria. The leaves are very nutritious since they contain up to 20% protein on a dry weight basis, in addition to appreciate amounts of vitamins and minerals. Like the corn and beans and must therefore be boiled before consumption.

— The cobs, cormels and leaves of Cojolans can be used as animal feed. In areas where the leaves are still green at harvest, there is a great potential for utilizing the leaves left after harvest for feeding animals. In order to minimize rejection to the animals and to improve palatability, it is advisable to turn the leaves into silage before feeding them to the animals.

PESTS AND DISEASES ✓

DISEASES

Taro leaf blight is caused by fungus Phytophthora Colocasiae. The disease appears on two leaves a water-soaked lesions which later spread to cause necrosis of most of the leaf.

Control: It can be controlled by the use of copper-based fungicides such as copper oxychloride and tribasic copper sulphate.

② The use of resistant varieties is also recommended. The disease does not attack banana P.T.O

Pathogen exists in the soil and attacks the roots, corns and cornels, causing them to rot. The leaves also yellow and wither.

Control: ① The disease is controlled by the use of resistant varieties.
② by planting disease-free material ~~and by~~
③ by crop rotation.
④ Treating the soil and planting material with Captan fungicide has sometimes proved effective.

Nematodes

Root-knot nematodes (Meloidogyne spp.) is caused galls and irregular swelling on the underground portions of cotyledons, resulting in stunting of the plant.

Control: ① Control is by planting nematode-free material.
② Planting material that is already infested should be treated with water at 40°C for 50 minutes.
③ Infested soil should be fumigated with a nematocide such as Mennagon or Heptanema.

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leaves and can decrease yield by 15-20%.
Control:- By the use of animal repellants
e.g. anemoll which has been used in the
Gambia.

COFFEE (Coffea sp) ✓ ORIGIN AND DISTRIBUTION

There are 2 varieties of coffee, Arabica and Robusta.

- (A) Arabica coffee (Coffea arabica L.) and
- (B) Robusta coffee (Coffea canephora Pierre
et Froehner).

Arabica coffee (Coffea arabica L.):-

(1) The centre of origin of Arabica coffee

(Coffea arabica L.) is in Ethiopia where it
grows wild in the forest. It was domesticated
only recently compared with most other
crops. It was cultivated, and a 'wild' was
first made from the roasted seeds in Yemen.
From Yemen, it was introduced to India and
Sri Lanka at the end of the 17th century, and
very soon afterwards to Java (Indonesia).

A single plant from Java was taken by the
Dutch to Basterdam in 1706 and from there,
coffee was taken to Suriname in 1718, to the
Philippines in 1740 and Hawaii in 1825.
From Suriname it was taken to Brazil in 1725.

By the middle of the 18th century coffee was
spread throughout the Caribbean and Central
and South America.

(2) Robusta coffee (Coffea canephora Pierre et Froehner):-
It occurs wild in African equatorial forest
from the west coast to Uganda, chiefly between
10°N and 5°S. Planting material was taken from
the Congo to Belgium. From there it was

sent to many botanical gardens in Europe. In 1900 it was introduced from Belgium to Java (Indonesia) where it proved very successful under the name of robusta coffee to distinguish it from arabica. In 1989 it was sent from Kenya to Singapore and Trinidad. Since then, robusta coffee has been widely distributed throughout the tropics where it is grown successfully at higher elevations, which are unsuited to arabica. It is now the most important species in tropical Africa and Asia.

AREA AND PRODUCTION

The area and production of green coffee in the world during 2007 were 14.5 million hectares and 10.2 million tonnes, respectively. In the same year, the area and production of green coffee in Africa were 0.4 million hectares and 5.8 million tonnes, respectively. Maximum production comes from Brazil. Other important coffee-producing countries are Colombia, Indonesia, Mexico, Côte d'Ivoire, Guatemala, India, Ethiopia, Uganda, the Philippines, Costa Rica, Ecuador, Kenya, Peru, Honduras, Zaïre, Cameroon and Madagascar.

BOTANY OF COFFEE

Coffea arabica is a small tree which grows up to 8m tall if it is not pruned. The plant has a short, stout tap root, rarely extending beyond 45 cm deep. There are 2 kinds of vegetative growth of branches: vertical and growth and horizontal growth. Vertical growth type branches only produce

Further vegetative growth while horizontal growth type branch can produce both reproductive growth and vegetative growth.

Each inflorescence usually consists of about 4 flowers, but not all 4 may develop. The flowers are white and sweetly scented. Each flower produces one fruit normally containing 2 seeds which are called beans. Each of the 2 beans remains enclosed in a tough membrane called the parchment. Fruit development is a long process and the ripening period, from flowering to maturity, is normally 8-9 months.

The seeds of *C. arabica* are 8.5-12.5 mm long while the seeds of *C. canephora* are about 8.5 mm long.

ECOLOGICAL REQUIREMENTS

Coffea arabica is a tropical tree. The ideal conditions for *arabica* coffee are mean temperature of 16-24°C and an annual rainfall of about 1,500-2,250 mm which is well distributed with a drier period of 2-3 months for the initiation of the flower buds. In low rainfall areas, coffee is grown with irrigation and heavy mulching. *Coffea canephora* is not as specific in its requirements as *C. arabica* and stands a wider range of adaptability. Its requirements for temperature and rainfall are the same as for *C. arabica* but it is more tolerant of adverse conditions and poor management. The best soils for coffee are deep, well-drained, fertile loams of laterite or volcanic origin with a reasonable humus content and slightly acidic (pH 5.5-6.0).

Heavy loams and clays are unsuitable because they are poorly aerated; sandy soils are equally undesirable because they dry out rapidly.

PROPAGATION ✓

Raising Seedlings

Coffee is usually propagated from seed, though it can be propagated vegetatively by budding, grafting or from stem cuttings.

The seed are sown in a nursery which should be prepared on level land with deep fertile soil, and be near a source of water. Arrangements for shade must be made before sowing. Seeds are sown either thickly with the intention of transplanting the seedlings at the cotyledon stage, again in the nursery, or thinly so that the seedlings are left undisturbed until they are ready to be transplanted directly to the field. Seedlings may also be raised in seed-bases or polythene bags. Germination is hastened by 4-8 weeks and seedlings are transplanted in the field at the 6-leaf stage, when 6-18 months old. Weak seedlings should be discarded.

Transplanting in the Field:

The land should be prepared well in advance of transplanting. All vegetation should be removed from the field and the planting holes should be dug at least 3 months before transplanting. The spacing should be 2.7m x 1.4m. At least 2-3 weeks before transplanting, the top soil of each planting hole is taken out and mixed with manure

and 30g of superphosphate. The holes are then refilled with this mixture and additional top soil if necessary.

Transplanting is done at the beginning of the rainy season. The holes are reopened immediately before transplanting. The seedlings should be carefully uprooted so that they carry some soil around their roots. Transplanting should also be done carefully so that the roots are not bent. Soon after transplanting, mulch should be placed around the seedlings, covering the seedlings with banana leaves after transplanting is also beneficial.

MANAGEMENT

1. Fertilization

In Africa, nitrogen is the main element that is applied to coffee although in other countries compound fertilizers containing N, P and K are applied. The full dose of nitrogen may either be applied only once or as split applications: the first at the beginning of the long rains, the second a month later and the third during the short rains.

Before applying fertilizer, all mulch should be removed. In young coffee, fertilizer can be spread in a ring around, but not very near to, each plant. In mature coffee, fertilizer is usually spread in a band along the inter-rows.

2. Weed control

Coffee is highly intolerant of competition with weeds. If the land is bare i.e. without any inter-crop or mulch, continuous cultivation keeps the weeds under control. This can be done with hand tools.

When the coffee plants are young, the direct application of paraquat may be very effective in controlling weeds.

3 Pruning

The coffee tree, if left unchecked, will grow as high as 8 m and can then only be harvested with difficulty. It is therefore essential to limit the height of the tree by pruning. The best time to prune is in the dormant period after harvesting and before the new flush of growth begins.

Harvesting

Coffee trees come into bearing 3-4 years after transplanting, are in full bearing at 6-8 years and may continue to bear for 50-60 years. The first mature 7-9 months after flowering. The fruit are collected by hand when they are mature, either by direct picking from the branches or after they have been shaken to the ground. The entire surface of a ripe berry is red. Over ripe berries (darkly) and under-ripe berries (green) are not good. As all the berries do not ripen at the same time, several pickings at 10-14 day intervals are needed.

PROCESSING OF COFFEE

There are 2 methods of processing, the ^①dry method and the ^②wet method.

In the dry method the mature berries are spread out thinly and dried in the sun without being pulped for about 15-25 days. They are hulled. Coffee beans processed in this way are inferior to those subjected

to wet processing. Most of Brazil's coffee is prepared in this way.

2. In the wet method, the mesocarp of the berry is removed in a pulping machine soon after picking (not longer than 24 hours after picking or they begin to ferment). Under- and over-ripe fruits are picked separately. After pulping, the seeds are washed and then passed onto the fermenting tank for about 12-24 hours. During fermentation the remains of the mesocarp and the mucilage which surrounds the parchment are removed and the aleurone, bright, greyish-blue colour of the seed is developed. The seeds are then dried on racks in the sun for a week or more, during which they are turned regularly and protected from rain. Finally the dried beans are hulled by a machine which removes the parchment. They are then polished to remove the silver skin. In Africa, this is the only method of processing used.

YIELDS

The turn-out of clean dry coffee from ripe cherry is 15-20%. Average yields of clean coffee beans vary between countries and estates. The average yield in Brazil is about 500 kg/ha and the average yield in East Africa is about 900 kg/ha. In Kenya a few estates have yielded up to 2000 kg/ha.

CROP PROTECTION DISEASES

The important diseases that attack coffee in Africa are coffee berry disease, El Guano disease and leaf rust.

1. Coffee berry disease:-

This disease is serious host in many African countries. The strain which is parasitic, attacks the berries in the green stage, causing brown sunken spot. Badly affected berries go completely brown to black, and the berries inside are destroyed.

Control:- Application of fungicides at flowering and early fruiting.

2. Elgon disease:-

The first symptom is blackening of the node and petiole, followed by the death of the branch above the node.

Control:- (i) Use of resistant varieties.

(ii) Regular spraying with fungicides.

3. Leaf rust:-

First of all yellow spots appear on the leaves. These enlarge rapidly, forming circular lesions which have orange pustules on the lower surface, later defoliation occurs.

Control:- (i) Use of resistant varieties.

(ii) Spraying with copper fungicides.

INSECT PESTS

1. Leaf miners:-

The eggs are laid on the upper surface of the leaves. Upon hatching, the larvae bore into the leaf and defoliation occurs.

Control:- Repeated spraying with recommended insecticide.

2. Coffee bugs:-

These pests suck the sugarcane and feed

becoming increasingly serious in sweet potatoes in Africa. The causal organism is a strain of the tobacco mosaic virus. Symptoms of the disease include small, malformed leaves and a considerably reduced yield of tuber.

Control

The removal and burning of infected plants is an effective control measure.

FIBRE CROPS COTTON (Gossypium spp.) ✓

ORIGIN AND DISTRIBUTION

The origin of the different species of Gossypium has been a subject of considerable controversy. The various species have originated in different centres. The wild but less diploid species probably evolved in South Africa and spread from there to the arid regions of Arabia, South-East Asia, Australia and America.

So far 31 species of cotton have been recognised, out of which only four species are cultivated; two diploid species, G. herbaceum and G. arboreum of the old world, and two tetraploid species, G. barbadense and G. hirsutum, of the New world.

The leading countries for cotton production

TOBACCO (*Nicotiana glauca* L.) ORIGIN AND DISTRIBUTION

Tobacco is native to tropical America, where it was used for chewing, smoking or as snuff. Long before it was discovered by Europeans. The Spanish carried tobacco from Mexico to Spain in 1558. From Florida it was introduced to England in 1585. The commercial production of tobacco was begun in Haiti by Spaniards in 1580 and in Trinidad in about 1595. The crop was introduced to Turkey in 1610, and to Virginia in 1612. It was widely spread to India and other Asian and Portuguese at the beginning of the 17th century.

The total area and production of cured tobacco leaves in the world during 1984 were 5.0 million hectares and 7.3 million tonnes, respectively. The important tobacco-producing countries in the world are China, the USA, India, Brazil, the USSR, Turkey and Zimbabwe.

The total area and production of cured tobacco leaves in 1984 in Africa were 0.34 million hectares and 0.36 million tonnes, respectively. In addition to Zimbabwe, Malawi, Tanzania, Kenya, Nigeria and Zaïre are important tobacco-producing countries in tropical Africa.

BOTANY OF TOBACCO

The genus Nicotiana belongs to the family Solanaceae. The genus contains over 50 species, of which only two, Nicotiana glauca and N. rustica are cultivated species.

Tobacco is normally grown as an annual, it is potentially a woody shrub - Woe, Short-lived perennial. The seedling has a single main tap root, but this is generally broken during transplanting.

The stem is unbranched, but after topping (removing of the inflorescence at a later stage) each axillary bud grows out to produce a branch or sucker. The stem bears large, simple, ovate leaves arranged spirally.

The leaves vary greatly in size, thickness, texture and in the prominence of their veins, depending upon variations in cultivar. They are about 50 cm long and sessile. The number of leaves, 20-30, is usually fairly constant in each cultivar.

The inflorescence is terminal on the stalk and multi-flowered. The anthers albicose and the stigma is receptive when the flower opens so that self-pollination is the rule. Up to 4% cross-pollination does occur because bees and other insects visit the flowers for nectar. The fruit is two-valved ovoid capsule, 1.5-2 cm long.

almost completely covered with calyx. It takes
cas longiturnally to release no more than
8000 seeds.

ECOLOGICAL REQUIREMENTS

Tobacco is one of the most adaptable of
tropical crops, and grows well in the subtropics
and even in temperate climates if there is a
frost-free period of 90-120 days from trans-
planting to harvesting.

The optimum mean temperatures for the grow-
ing season are 25-30°C. Tobacco requires a
moderate rainfall of around 1400 mm during
the 3-4 month growing season, followed by dry
weather when the leaves are maturing. Continuous
rain during the growing season leads to disease
and thin, light-weight leaves. A prolonged dry
period during ripening causes secondary growth
and a deficiency of gum on the leaves.

In general the crop requires a well-drained,
moderately fertile, well aerated soil with a
high moisture-holding capacity. It does not
tolerate waterlogging and strongly acidic and
alkaline soils are also not suitable.

CULTIVATION OF TOBACCO

The cultivation of tobacco consists of raising seed-
lings from seed in nursery beds and transplanting
the seedlings to the field.

RAISING SEEDLINGS

Because of their minute size, germinating

Tobacco seeds are very delicate. This means the seedlings have to be raised with great care. Raising seedlings in a nursery facilitates supervision of their growth and in particular allows the early detection and easier control of pests and diseases. It is also much easier to keep a seed bed free of weeds.

The site should be carefully chosen to allow easy supervision. The site must be near a source of water to facilitate irrigation. The sterilization and fumigation of nursery beds have become common practice in most tobacco areas. The aims are to control weeds by killing their seed, and to destroy nematodes and other disease organisms. Steaming is a very effective method of sterilization, but it is very expensive and time consuming. Fertilizer is applied after any sterilization process. If early growth is slow or the plants are very yellow, nitrogen in nitrate form can be applied as a top dressing.

It is essential to keep the surface of the bed moist from the time of sowing until the seedlings have their roots well down. The first 10 days are most critical and the seed bed must be watered regularly. In addition, the bed are covered with either a plastic sheet, a cloth or chopped grass mulch to protect the seedlings from cold, wind moisture loss. The young seedlings grow through the mulch without any adjustment to their density. The transplants

of cover is decreased gradually. Besides weeding and the regulation of cover, the other operations necessary during seedling growth, are weeding and thinning.

As the seedlings approach transplantable size they need to be prepared for the shock of transplanting in the field. Where thinning is done during the rainy season, the cover has to be completely removed at an early stage so as to accustom the plants to light.

TRANSPLANTING

Height is a convenient basis for judging when a seedling is ready for transplanting. Normally the height should be about 10-15 cm from the root crown to the bud.

The field into which the seedlings are to be transplanted should be ready to receive the seedlings while a fine till is not necessary, it is important to have the soil fairly well broken down so that there are no large clods. In some places planting is done on ridges.

Transplanting should normally be done in the afternoon and there should be sufficient moisture in the soil at the time of transplanting. The seedlings are planted in rows 60-100 cm apart, the distance between the plants in a row varies from 30-60 cm depending on the size of cultivar and soil fertility. If soil moisture is inadequate at the time of transplanting, a small amount of water should be applied to each plant just

transplanted. Sometimes it is advisable to cover the transplants with dried grass or leaves during the daytime for 2-3 days until they are established.

CROP MANAGEMENT

(a) Fertilization:-

Fertilizer management is of critical importance in tobacco production as it makes the quality of the leaf. Excessive nitrogen produces a strong flavour and an undesirable dark green leaf that cures with dark colours. For other types of tobacco, the fertilizer should contain 18 parts of tobacco, the fertilizer should contain 18 parts on the roots 1:2:2 or 1:2:3. Fertilizers should be banded 7.5-10 cm on one or both sides of the seedlings and about 5 cm below the soil surface for best results. This practice prevents fertilizer injury to the seedlings.

(b) Weed Control

Under no circumstances should weeds be allowed to grow in the tobacco field as they not only greatly reduce yields but also reduce the quality of the cured leaf and hence the market price. Mixtures of MCPA and TCA applied 15-20 days before transplanting have been found very effective in controlling weeds.

HARVESTING

There are two methods of harvesting which are:-
1) The leaves may be cut off individually so that all are gathered at the same stage of ripeness, or
2) The whole stalk may be cut and the leaves not removed from it until they are fully cured.

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Ripeness is indicated by a lightening of the colour of the blade, a shortening of the midrib and an angle approximately 90° between the stem and the base of the midrib.

In USA and most other countries, the air-cured and fire-cured and most cigar tobacco (except the wrapper) are harvested whole by cutting the stem near ground level when the largest number of the leaves are at the correct stage of ripeness. This is usually 40-55 days after topping. Topping is the operation that takes place when the flower buds are formed, the inflorescence and the topmost leaves are removed. The leaves are cured on the stems, which are tied to poles, and the leaves are rolled before they are transferred to the curing barn.

CURING

Curing may be described as the process by which the harvested tobacco leaf is made ready for the market. It is essentially a drying process whereby most of the moisture in the harvested leaf is removed. The process of drying is conducted in such a way as to produce certain well-defined and desirable qualities in different types of tobacco. Depending on the method of harvested and the maturity of the leaf at the time of harvest, there are 4 principal methods of curing:—

- ① flue-curing ② air curing,
- ③ fire-curing and ④ sun-curing.

very short description of different curing processes are given in the following.

① Fire-curing, tobacco is cured in the oven until almost all the starch has been converted to sugar but before any appreciable amount of the sugar has been reduced further. The desired stage is conveniently denoted by a colour change; once it has been reached, the leaf is fixed in that stage by the use of artificial heat which dries it out and kills it.

Flue-curing is the shortest process (3½-6 days).
2. Sun-curing is done by exposing the leaves to the sun but not under any precise control. For the first 2-3 days the leaves are air-cure and then they are put in heaps for 24-36 hours to ferment, before they are exposed to the sun. Sun-curing intermediate (2-4) weeks.

3. Air-curing, being a longer process, keeps the leaf alive until the initial sugar content has been completely oxidized away. It may be under fairly good control or no control at all. Curing is done under normal atmospheric conditions in a wooden or grass barn. Air-curing the longest (3-6 weeks).

4. Fire-curing is usually done in a barn. The leaves are hung for 4-7 days to yellow and are then smoked for 12 hours by burning dried, grass, husks, twigs etc., in a pit on the floor. This produces the creosotic and althnic aroma of fire-cured tobacco.

YIELDS

Very variable yields of tobacco are obtained. In Africa, yields of cured leaves vary from 400-2000 kg/ha depending on the standard of husbandry. The highest yields in tropical Africa are obtained in Kenya and Zimbabwe where the average yield is more than 2000 kg/ha. The average yield in the United States and France is about 3400 kg/ha.

ECONOMIC IMPORTANCE OF TOBACCO

Tobacco is an important cash crop throughout the world.

Tobacco is consumed because of its stimulating and slightly narcotic effects and for its aroma. The stimulating and slightly narcotic effects of smoking, sniffing or chewing the dried, cured and fermented leaves of tobacco have led to the almost universal use of tobacco.

It provides a major source of revenue to the government.

It is smoked in the form of cigarettes, cigars and pipe tobacco.

- Cigarettes are made mainly from fire-cured tobacco.
- Cigars are made from air-cured leaf
- Pipe tobacco is usually made from blended fire-cured and air-cured mixture.
- Snuff is prepared by grinding up air-cured and fire-cured leaves.

PESTS AND DISEASES

Pests of Tobacco.

The insect pests that damage tobacco are cutworms, wireworms, aphids and thrips etc.

1. Cutworms (*Agrotis ypsilon*) cause damage in ear and central Africa. They feed only at night.

2. Wireworms feed on roots and organic matter and readily attack the roots of newly transplanted tobacco.

Control: - Parathion or diazinon can be used to ~~control~~ this pest.

3. Aphids feed on tobacco. They have a remarkable reproductive ability and are virus-transmitting agents. They are seen in colonies on the under surface of young leaves and tender plant parts. The nymphs and adults feed on leaves by sucking the cell sap. The leaves become yellow and crumple. The quality and yield of tobacco are badly affected.

Control: - Spray malathion 50 EC in 500 litre of water.

4. Thrips (*Thrips tabaci*) are tiny, slender insects. The eggs are laid on the leaf epidermis and both nymphs and adults feed on the leaf.

Control: - Control measures are similar to those for aphids.

5. Tobacco caterpillar (*Spodoptera litura*) : It is a pest of tobacco in seed nurseries. When the seedlings are small the caterpillars cut the tender stem just like the cutworms. They also feed on the tender leaves of the grown up seedlings.

Control: - Spray one litre of malathion 50 EC in 625

litters of water/ice and repeat at 2-3 weeks intervals. Other pests of tobacco are stem borer, saw leafper, ground beetle, mole cricket, whitefly, Green caterpillar, indigo ~~cater~~ caterpillar, hairy caterpillar.

Diseases of Tobacco

The main diseases of tobacco plants in Africa are damping off, frog eye, brown spot, anthracnose, Granville wilt, mosaic and rosette.

1. Damping off, caused by Pythium spp, and Guthrium solani, occurs in nurseries. The stems are attacked at soil level. The plants fall over and the leaves then rot into a shapeless slung mass.

Control: affected patches and surrounding areas may be dusted with copper fungicide.

2. Frog-eye is caused by Peridermium in cotininae. Circular spots with white or pale brown centre and narrow dark brown margins are produced on the leaves.

These brown margins are produced by Birdenavia Control: Normally, a weekly spraying of Bordeaux mixture or a proprietary copper compound will provide adequate control.

3. Brown spot is caused by Alternaria longipes. These brown spots with concentric rings can be distinguished from frog-eye by the presence of blackish patches on the stem, midribs and pedicels.

Control: Copper compound and maneb have been used with some success.

1. Anthera is caused by Colletotrichum Tobacco.
usually occurs in nurseries. In later - soaked patches
are formed on the lower leaves.

Control: - Control can be effected with zinc or
thiram.

5. Graville wilt is caused by Pseudomonas solanacearum,
caused delay of the roots, followed by withering.

Control: Copper-based sprays or dusts provide
effective control both in the nursery and in the field.

6. Mosaic is a virus disease transmitted by mechanical
contact. The common symptom is dark green and
yellowing mottling, but the disease can also cause
spotting and scorching.

Control: - Control is through prevention and the
destruction of all old tobacco debris.

7. Rosette is a virus disease transmitted by aphids
(Myzus persicae); the symptoms are stunting
and the production of many small leaves.

Control: Early planting to enable good growth
before a large build-up of aphids occurs.

~~Other diseases~~

① Green Mosaic: The virus attacks the plant, causing stunted growth, chlorosis or discolorated plants.

Control: Good crop rotation helps combat the disease.

② I am Nematode (Scutellonema bradyi) and the root-knot nematode (Meloidogyne spp) and attack you. The former causes damage to the meristematic region just beneath the tuber skin, so that affected tubers perform very poorly if used as planting material.

Control: Nematodes can be controlled by effective crop rotations.

③ Occasionally, nematicides may be used to fumigate the soil.

SWEET POTATOES (*Ipomoea batatas* L.)

ORIGIN AND DISTRIBUTION

Sweet Potato originated in Central America, and the north western part of South America. It was cultivated there for several centuries before it was introduced to Europe in the sixteenth century. It probably came to Africa from Europe during the voyages of discovery.

Today Sweet potatoes are grown in most parts of the tropics and the warmer temperate regions. Asia produces over 90% of the world's sweet potatoes, with China alone accounting

for over 85% of the world production. Other important producing countries on a world basis are Vietnam and Indonesia.

Africa accounts for only about 5% of world production of sweet potatoes.

PLANT OF SWEET POTATO

The sweet potato is a dicotyledonous herbaceous plant with a creeping growth habit. When propagated by vine (stem) cutting, the growth of the plant is marked by elaborate root growth during the first two months, extensive shoot enlargement in the third and fourth months and vigorous tuberculing for the rest of the season.

Sweet potato plants growing from vine cuttings develop a fibrous root system which penetrates deep into the soil.

A sweet potato plant produces several tubers. Each tuber is attached to the stem base by means of a stout stalk. Tubers may be spherical, cylindrical or oval in shape. The skin of the tuber may be whitish, brownish, yellowish or purplish, while the tuber flesh may be creamy white, yellowish or purplish.

The stem of the sweet potato plant is thin, cylindrical, herbaceous and creeping. Younger stem portions are hairy but the more mature parts are devoid of hairs.

The leaves are simple and arranged spirally on the stem. The petiole is readily long. The lamina is rhombic-lanceolate, and may be green or purplish in colour.

Flowers are borne on gynose inflorescences or solitary. There are five united sepals and five purplish petals which are united together to form a funnel-shaped structure. Flowering is promoted by short-day conditions. Each flower is open for less than 12 hours before it wilt and dies.

The fruit is a capsule 5-8 mm in diameter. The seed is small (about 3 mm long) and black. The testa is hard and impervious, so that stratification is often required for the seeds to germinate.

ECOLOGICAL REQUIREMENTS

Sweet potatoes are a warm-weather crop and grow best at temperatures above 24°C . The crop does best in abundant sunshine, although both tuber formation and flowering are promoted by short day length.

A well-distributed rainfall of 750-1000 mm per annum is best for the crop. Drought is particularly harmful to the crop during the first six weeks after planting and at the time of tuber initiation. Moisture in the form of rainfall or irrigation water must be available during the dry (non-cropping) season in order to permit the maintenance of plants whose vines will be used as planting material in the following season.

The best soil for sweet potatoes is sandy loam

Poorly drained, poorly aerated or saline soils tend to retard tuber development. Growth is best at a pH of about 6; alkaline soils result in poor yield.

LAND PREPARATION

Land preparation for planting sweet potatoes is similar to that for planting cassava ~~and~~ yams. Mounds, ridges and planting on the flat are all common place. Planting on the flat yield lower than the others and are more difficult to harvest.

PLANTING MATERIAL

The best planting material for sweet potatoes is the tip, the top 30 cm or so of the vine. Other parts of the vine may all be used if there is a shortage of tip portions. Pieces of the tuber or small intact tubers may sometimes be used for the commercial propagation of sweet potatoes, but this practice results in low yield and utilizes tuber material which could otherwise have been used for food. Moreover, tuber material, unlike vine material, is able to transmit soil-borne diseases to the new crop.

CROP ESTABLISHMENT

Sweet potatoes are normally planted at a within-row spacing of 30-25-30 cm ~~by~~ ridged or rows that are 60-75 cm apart.

Usually one vine is planted on each spot, but when mounds are used, 2 or 3 vines may be planted on various portions of the same mound. Where sweet potatoes are intercropped the spacing between the plants will depend on the number and density of the other crops. It is best to plant sweet potatoes early in the rainy season so that the crop can make adequate growth and mature before the onset of the dry season.

CROP MANAGEMENT

1. Fertilization:

Even though sweet potatoes may give reasonable yields on relatively poor soils, the crop usually responds to fertilizer application. At the level of the traditional farmers, farmyard manure or

compost may be used.

For chemical fertilizers at 6:4:15 NPK fertilizer applied at 50-1120 kg/ha may be generally recommended. The exact type and dosage of fertilizer used will depend on the nature of the soil, but excessive nitrogen fertilizer should be avoided since it delays tuber formation and promotes shoot growth at the expense of tuber growth.

Fertilizer applied to sweet potatoes should normally be split into two doses. The first

dose is applied immediately after planting either dissolved in water and sprinkled on the plants or as a band or spot application beside the plant. The second dose is applied at five or six weeks after planting.

2. Weed control

Sweet potato vines grow rapidly and compete effectively with weeds. For this reason, weed control measures are only necessary for the first two months after planting. Thereafter the crop, if planted at the correct spacing, covers the ground effectively and smothers most weeds.

In traditional practice, weed control with hand tools is done about 4 weeks after planting.

Herbicides used control in sweet potato plots is not widespread in Africa, although it is common practice in some other parts of the world. Herbicides that have been found effective include alachlor and Amiben (Chloramben), each at about 3.5 kg/ha.

HARVESTING

Most sweet potato cultivars are mature for harvesting ~~five~~ five months after planting. Harvesting too early results in low yields while an unduly delayed harvest may cause the tubers to become fibrous. Delaying

harvest for too long also encourages the build-up of the sweet potato weevil and attacks the tubers.

In most parts of Africa, sweet potatoes are harvested manually by means of sharpened stakes, metal rods or machetes. Care should be taken to avoid tuber wounding during the harvesting operation.

Mechanical harvesters for sweet potatoes are in use in various parts of the world. For machine first remove the vines and then dig out the tubers.

YIELDS

Sweet potato yields in Africa average approximately 5 t/ha. These are rather low yields compared with average yields of 16 t/ha in Asia and 10 t/ha in South America. The main reason for these low yields in Africa are poor crop management and the use of low-yielding cultivars. In some restricted locations and at experimental stations, yields as high as 20 t/ha are also being regularly obtained.

STORAGE

After harvesting but before storage, the tubers are subjected to curing, a process which promotes the healing of wounds inflicted during

harvesting, and to dryness the tuber skin curing is best done by subjected the freshly harvested tubers to a temperature of $27-29.5^{\circ}\text{C}$ and relatively humidity of $85-90\%$ for 4-7 days. This is usually done in a well-ventilated storehouse. After curing, the best conditions for storing sweet potatoes are a temperature of $13-16^{\circ}\text{C}$ and a relative humidity of $85-90\%$. Most producers in Africa supply store than sweet potatoes under ambient conditions baskets, in underground pits, or on platforms for more prolonged storage of sweet potatoes, the lowering of temperatures with refrigeration is essential.

ECONOMIC IMPORTANCE

- The fresh tuber of sweet potato consists of about 60-73% water, 15-22.5% starch, 1-2% protein and 1-2% sugars.
- The tuber also contains significant amounts of vitamins and minerals. The vitamins are mainly carotene, ascorbic acid and the B complex contained in the tuber. The important minerals contained are potassium, sodium, chloride, phosphorus, calcium.
- Most sweet potatoes produced in Africa are consumed by boiling, baking, frying or roasting the unprocessed tubers. Whenever possible, it is best to cook the tubers before peeling since this permits some of

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- Vitamins and minerals which are present in the peel permeate into the edible flesh.
 - Sweet potato flour is made by drying thin slices of the tuber and then milling the dried slices.
 - Sweet potato leaves is used for human consumption. The young leaves are particularly rich in protein, minerals and vitamins. They are usually boiled before being eaten.
 - Both the tuber and shoot can serve as livestock feed either in the fresh form or in the form of chips.

PESTS AND DISEASES

Pests of Sweet potatoes are: —

- (a) Sweet potato weevil's (*Ceyletus* spp) are one of the most serious pests of sweet potato. The female lays eggs in small pockets which it makes at the base of the stem and in the tuber. The resulting larvae then tunnel extensively through the tuber, feeding on it and depositing frass within it. This also promotes the rotting of the tubers.

Control

The following control measures can be adopted:

- (1) Use of resistant cultivars
- (2) Use of insecticides such as 0.1% carbaryl, 0.1% fenitrothion, 0.1% fenitrothion or 0.1% kdp.

- ③ effective quarantine to prevent the spread of the weevil in transported plant material.
- ④ crop rotation with non-susceptible crops.
- ⑤ planting weevil-free material.

Harle maths (these convulvulids) are a serious pest of sweet potatoes in Africa. The moth lays its eggs in soil, and the larvae feed extensively on sweet potato leaves.

Control:

Control is by spraying with suitable insecticides.

Diseases of sweet potatoes

① Scabs of sweet potatoes are: -

- (a) Black rot: - is caused by the fungus Cercosypha funebrata. It may attack the plant in the field, or the tuber during storage. It causes a blackening of the tuber and underground stem portions. Infected tubers produce same toxic which cannot be removed by boiling.

Control:

The disease is controlled by:-

- ① the use of resistant varieties
- ② crop rotation
- ③ by washing the tubers with borax or another fungicide before storage.

② Mosaic - is a virus disease which is